Convergence and Error Analysis of the CROWN-Coupled Masked Dictionary Learning Framework for Image Inpainting

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Abstract

We present a self-contained mathematical analysis of the *Masked Dictionary Learning with CROWN (Confidence, Regime, Non-local) Coupling* framework for image inpainting under arbitrary observation masks. The framework combines three optimisation primitives: (i) a masked rank-one Alternating Least Squares (ALS) dictionary update that side-steps the ill-defined SVD on incomplete data, (ii) a confidence-weighted Orthogonal Matching Pursuit followed by a non-local centralised ℓ_1 refinement (NSR), and (iii) a coarse-to-fine pyramid that supplies a non-trivial warm start to the non-convex outer loop. We prove four theorems that together establish the algorithm's convergence and quantify its reconstruction accuracy. Specifically, we show that the masked ALS subproblem is monotonically non-increasing and converges to a coordinate-wise stationary point under a verifiable atom-utilisation condition; that the outer loop inherits this descent property modulo the standard greedy-OMP approximation; that the non-local centralised refinement admits a slow-rate Lasso oracle inequality whose error is governed by a *Self-Similarity Index* $\mathrm{SSI}_i := \|\boldsymbol{\alpha}_i^\star - \boldsymbol{\beta}_i\|_1$; and that the multi-scale warm start strictly removes the zero-information plateau of the loss surface that traps a single-scale K-SVD on contiguous holes larger than the patch.

Index Terms

Image inpainting, dictionary learning, K-SVD, sparse coding, masked OMP, non-local self-similarity, alternating least squares, block-coordinate descent, convergence analysis, multi-scale optimisation.

I. Introduction

Patch-based dictionary learning methods such as K-SVD [1] provide state-of-the-art reconstruction quality on a wide class of image restoration problems, but their use in inpainting is hampered by a structural obstruction: the SVD step in the dictionary update is not defined when the leave-one-out error matrix has unobserved entries, and patches lying entirely inside the hole carry no observation and therefore no gradient. The CROWN-Inpaint framework addresses both issues simultaneously through three coupled mechanisms:

1) A *masked ALS rank-one update* that replaces the SVD step by closed-form scalar weighted least squares restricted to observed entries.
2) A *confidence-weighted, non-local centralised* sparse-coding step that fuses an OMP support-selection with an ℓ_1 refinement centered at the weighted mean code of the patch's nearest neighbours.
3) A *coarse-to-fine pyramid* that initialises the dictionary and codes from a downsampled scale at which every hole patch sees at least one observed pixel.

The aim of the present paper is to provide rigorous theoretical guarantees for each of these mechanisms in isolation, and to compose them into an end-to-end convergence statement for the full pipeline.

II. Problem Setup and Standing Notation

Let $\boldsymbol{x}_1, \ldots, \boldsymbol{x}_N \in \mathbb{R}^n$ be vectorised image patches stacked column-wise into $X \in \mathbb{R}^{n\times N}$, and let $M_i \in \{0,1\}^{n\times n}$ for $i \in [N]$ be diagonal binary mask matrices with $(M_i)_{jj} = 1$ iff coordinate j of patch i is observed. Denote the

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observed coordinate set by $\Omega_i := \{j : (M_i)_{jj} = 1\}$. Let $D = [\boldsymbol{d}_1, \ldots, \boldsymbol{d}_K] \in \mathbb{R}^{n\times K}$ be a unit-norm dictionary ($\|\boldsymbol{d}_k\|_2 = 1$), and $A = [\boldsymbol{\alpha}_1, \ldots, \boldsymbol{\alpha}_N] \in \mathbb{R}^{K\times N}$ the matrix of sparse codes. The *masked dictionary-learning energy* is

$$\boxed{\mathcal{L}(D, A) \;=\; \sum_{i=1}^{N} \|M_i\,(\boldsymbol{x}_i - D\boldsymbol{\alpha}_i)\|_2^2, \quad \text{s.t. } \|\boldsymbol{\alpha}_i\|_0 \le s,\ \|\boldsymbol{d}_k\|_2 = 1.} \tag{1}$$

For a fixed atom index $k \in [K]$, the *active set* of signals using atom k is

$$\mathcal{S}_k \;:=\; \{\, i \in [N] \,:\, A_{ki} \neq 0 \,\}. \tag{2}$$

We will reserve the symbol $\boldsymbol{h} \in \mathbb{R}^{|\mathcal{S}_k|}$ for the restriction of the k-th coefficient row $A_{k,:}$ to $\mathcal{S}_k$.

Definition 1 (Leave-one-out residual)**.** For $k \in [K]$ and $i \in \mathcal{S}_k$, the leave-one-out error of atom k on signal i is

$$\boldsymbol{e}_{k,i} \;:=\; \boldsymbol{x}_i - \sum_{\ell\neq k} \boldsymbol{d}_\ell\, A_{\ell i} \;=\; (\boldsymbol{x}_i - D\boldsymbol{\alpha}_i) + \boldsymbol{d}_k\, A_{ki}. \tag{3}$$

The identity (3) is the algebraic reason ALS can update $(\boldsymbol{d}_k, \boldsymbol{h})$ holding $\{\boldsymbol{d}_\ell, A_{\ell,:}\}_{\ell\neq k}$ fixed.

Assumption 1 (Atom utilisation)**.** For every $k \in [K]$, the active set $\mathcal{S}_k$ is non-empty and

$$\exists\, j \in [n] \;:\; \sum_{i\in\mathcal{S}_k} (M_i)_{jj}\, A_{ki}^2 \;>\; 0, \qquad \exists\, i \in \mathcal{S}_k \;:\; \sum_{j\in[n]} (M_i)_{jj}\, d_{k,j}^2 \;>\; 0. \tag{4}$$

Assumption 1 is the masked counterpart of the standard atom-utilisation hypothesis in K-SVD analysis [1]; it fails only when an atom is either unused everywhere or completely "hidden" by the masks of all signals that use it. The implementation re-initialises any such atom from a unit Gaussian (see `masked_ksvd.py:401`), so the assumption holds inductively.

III. Convergence of the Masked ALS Dictionary Update

For a fixed atom index k we consider the rank-one ALS subproblem

$$\min_{\boldsymbol{d}_k\in\mathbb{R}^n,\ \boldsymbol{h}\in\mathbb{R}^{|\mathcal{S}_k|}} \Phi_k(\boldsymbol{d}_k, \boldsymbol{h}) \;:=\; \sum_{i\in\mathcal{S}_k} \|M_i\,(\boldsymbol{e}_{k,i} - \boldsymbol{d}_k\, h_{k,i})\|_2^2. \tag{5}$$

A. Closed-form scalar updates

Lemma 1 (Stage A: scalar update of $h_{k,i}$)**.** *For fixed $\boldsymbol{d}_k$, the unique minimiser of Φ_k in $h_{k,i}$ is*

$$h_{k,i}^\star \;=\; \frac{\sum_{j=1}^n (M_i)_{jj}\, d_{k,j}\, e_{k,i,j}}{\sum_{j=1}^n (M_i)_{jj}\, d_{k,j}^2}, \quad i \in \mathcal{S}_k, \tag{6}$$

provided the denominator is non-zero. Each $h_{k,i}^\star$ is independent across $i \in \mathcal{S}_k$.

Proof: The map $h \mapsto \Phi_k(\boldsymbol{d}_k, h)$ is separable across i:

$$\Phi_k(\boldsymbol{d}_k, \boldsymbol{h}) = \sum_{i\in\mathcal{S}_k} \sum_{j=1}^n (M_i)_{jj} \big(e_{k,i,j} - d_{k,j}\, h_{k,i}\big)^2.$$

For each i, the function $\phi_i(h) := \sum_j (M_i)_{jj}(e_{k,i,j} - d_{k,j}h)^2$ is a convex scalar quadratic with leading coefficient $a_i := \sum_j (M_i)_{jj} d_{k,j}^2 \ge 0$. Setting $\phi_i'(h) = 0$ yields (6). Strict convexity, hence uniqueness, holds whenever $a_i > 0$, which is guaranteed by Assumption 1. ∎

Lemma 2 (Stage B: scalar update of $d_{k,j}$)**.** *For fixed $\boldsymbol{h}$, the unique minimiser of Φ_k in $d_{k,j}$ is*

$$d_{k,j}^\star \;=\; \frac{\sum_{i\in\mathcal{S}_k} (M_i)_{jj}\, h_{k,i}\, e_{k,i,j}}{\sum_{i\in\mathcal{S}_k} (M_i)_{jj}\, h_{k,i}^2}, \quad j \in [n], \tag{7}$$

provided the denominator is non-zero. Each $d_{k,j}^\star$ is independent across j.

Proof: Symmetric to Lemma 1, exchanging the roles of $\boldsymbol{d}_k$ and $\boldsymbol{h}$. ∎

The closed forms (6)–(7) are exactly the update rules implemented in `masked_ksvd_update_atom` (`masked_ksvd.py`
where Stage A is vectorised over $i \in \mathcal{S}_k$ via the broadcast outer product $d_k[:,\text{None}] \odot M_{\mathcal{S}_k}$ and Stage B over $j \in [n]$ via $h_k[\text{None},:] \odot M_{\mathcal{S}_k}$.

B. Norm-balancing invariance

Lemma 3 (Scale invariance of Φ_k)**.** *For any $\rho > 0$, the rescaling $T_\rho(\boldsymbol{d}_k, \boldsymbol{h}) := (\boldsymbol{d}_k/\rho,\ \boldsymbol{h}\,\rho)$ preserves the rank-one product $\boldsymbol{d}_k\boldsymbol{h}^\top$ and therefore $\Phi_k\big(T_\rho(\boldsymbol{d}_k, \boldsymbol{h})\big) = \Phi_k(\boldsymbol{d}_k, \boldsymbol{h})$.*

Proof: $(\boldsymbol{d}_k/\rho)(\boldsymbol{h}\,\rho)^\top = \boldsymbol{d}_k\boldsymbol{h}^\top$, and Φ_k depends on $(\boldsymbol{d}_k, \boldsymbol{h})$ only through $\boldsymbol{d}_k\boldsymbol{h}^\top$. ■

The post-ALS normalisation $\boldsymbol{d}_k \leftarrow \boldsymbol{d}_k/\,\|\boldsymbol{d}_k\|_2\,,\ \boldsymbol{h} \leftarrow \boldsymbol{h}\,\|\boldsymbol{d}_k\|_2$ (implemented at `masked_ksvd.py:395--398`) is therefore *cost-neutral*; its sole purpose is to maintain the unit-norm constraint required by OMP atom-selection in subsequent iterations.

C. Conditioning of the masked subproblems

The Hessian of $h \mapsto \Phi_k(\boldsymbol{d}_k, h)$ is the diagonal matrix $H^{(h)} = \mathrm{diag}(a_1, \ldots, a_{|\mathcal{S}_k|})$ with $a_i = \sum_{j\in\Omega_i} d_{k,j}^2 \in [0, 1]$. Hence

$$\kappa\big(H^{(h)}\big) \;=\; \frac{\max_{i\in\mathcal{S}_k} a_i}{\min_{i\in\mathcal{S}_k} a_i} \;=\; \frac{\max_{i\in\mathcal{S}_k} \sum_{j\in\Omega_i} d_{k,j}^2}{\min_{i\in\mathcal{S}_k} \sum_{j\in\Omega_i} d_{k,j}^2}. \tag{8}$$

The condition number is bounded whenever no signal $i \in \mathcal{S}_k$ has its observation set Ω_i *disjoint from the energetic support* of $\boldsymbol{d}_k$. Symmetrically, the Hessian for $d_{k,j}$ is the scalar $\sum_{i\in\mathcal{S}_k}(M_i)_{jj}h_{k,i}^2$, which vanishes only at coordinates never observed across $\mathcal{S}_k$ — but those coordinates contribute zero to $\mathcal{L}$ and are therefore harmless. The mask thus acts as a coordinate-wise weighting of an otherwise standard rank-one ALS; Assumption 1 suffices for well-posedness.

D. Monotone descent of masked ALS

Theorem 4 (Masked ALS descent)**.** *Let $(\boldsymbol{d}_k^{(t)}, \boldsymbol{h}^{(t)})$ be the iterates produced by alternating* (6) *and* (7)*, followed by the norm-balancing of Lemma 3, with $\boldsymbol{d}_k^{(0)} \neq 0$ and $\mathcal{S}_k \neq \emptyset$. Under Assumption 1,*

$$\Phi_k\big(\boldsymbol{d}_k^{(t+1)}, \boldsymbol{h}^{(t+1)}\big) \;\le\; \Phi_k\big(\boldsymbol{d}_k^{(t+1)}, \boldsymbol{h}^{(t)}\big) \;\le\; \Phi_k\big(\boldsymbol{d}_k^{(t)}, \boldsymbol{h}^{(t)}\big), \quad t \ge 0, \tag{9}$$

so the energy sequence is monotone non-increasing and converges to a limit $\Phi_k^\infty \ge 0$. Every accumulation point $(\boldsymbol{d}_k^\star, \boldsymbol{h}^\star)$ of $\{(\boldsymbol{d}_k^{(t)}, \boldsymbol{h}^{(t)})\}$ is a coordinate-wise (Nash) stationary point of (5)*, i.e. it satisfies* (6)–(7)*.*

Proof: **Step 1 (block monotonicity).** By Lemma 1, the update $\boldsymbol{h}^{(t+1)} = \arg\min_{\boldsymbol{h}} \Phi_k(\boldsymbol{d}_k^{(t)}, \boldsymbol{h})$ achieves the *global* minimum of Φ_k in its second argument with the first fixed; hence $\Phi_k(\boldsymbol{d}_k^{(t)}, \boldsymbol{h}^{(t+1)}) \le \Phi_k(\boldsymbol{d}_k^{(t)}, \boldsymbol{h}^{(t)})$. By Lemma 2 the symmetric statement holds for the $\boldsymbol{d}_k$-update, yielding the second inequality of (9).

Step 2 (norm balancing). The post-update normalisation is the operator T_ρ of Lemma 3 with $\rho = \|\boldsymbol{d}_k\|_2$. Since Φ_k is invariant under T_ρ, normalisation neither increases nor decreases Φ_k.

Step 3 (convergence of the energy). The sequence $\Phi_k^{(t)} := \Phi_k(\boldsymbol{d}_k^{(t)}, \boldsymbol{h}^{(t)})$ is non-increasing and bounded below by 0, hence it converges to a finite limit $\Phi_k^\infty \ge 0$.

Step 4 (boundedness of iterates). After normalisation, $\boldsymbol{d}_k^{(t)}$ lies on the unit sphere S^{n-1}, which is compact. From (6), $\big\|\boldsymbol{h}^{(t)}\big\|_\infty \le \|E_{k,\mathcal{S}_k}\|_\infty \,/\, \min_i a_i^{(t)}$, and Step 1 of Section III-C gives a uniform lower bound on $\min_i a_i^{(t)}$ under Assumption 1; hence $\boldsymbol{h}^{(t)}$ is also bounded.

Step 5 (stationarity at accumulation points). By Bolzano–Weierstrass, the bounded sequence $\{(\boldsymbol{d}_k^{(t)}, \boldsymbol{h}^{(t)})\}$ has at least one accumulation point $(\boldsymbol{d}_k^\star, \boldsymbol{h}^\star)$. Since the mappings in (6)–(7) are continuous on the open domain $\{a_i > 0\} \cap \{\sum_i (M_i)_{jj}h_{k,i}^2 > 0\}$, passing to the limit along a convergent subsequence yields $\boldsymbol{h}^\star = \arg\min_{\boldsymbol{h}} \Phi_k(\boldsymbol{d}_k^\star, \boldsymbol{h})$ and $\boldsymbol{d}_k^\star = \arg\min_{\boldsymbol{d}_k} \Phi_k(\boldsymbol{d}_k, \boldsymbol{h}^\star)$, the coordinate-wise (block-Nash) stationarity conditions. ■

Remark 1 (Relation to Gauss–Seidel BCD)**.** Theorem 4 is a specialisation of Tseng's block-coordinate descent theorem [7] to the two-block setting where each block sub-problem is strictly convex with a unique minimiser (Lemmas 1–2). The two-block case automatically satisfies the regularity conditions of [7], so no additional hypothesis is needed.

IV. Descent of the Outer K-SVD Loop

We now lift the per-atom analysis to the full masked dictionary-learning energy (1).

Lemma 5 (Atom-decomposition of $\mathcal{L}$)**.** *For any $k \in [K]$, the masked loss decomposes as*

$$\mathcal{L}(D, A) \;=\; \Phi_k(\boldsymbol{d}_k, \boldsymbol{h}) \;+\; \mathcal{C}_k, \qquad \mathcal{C}_k \;:=\; \sum_{i \notin \mathcal{S}_k} \|M_i(\boldsymbol{x}_i - D\boldsymbol{\alpha}_i)\|_2^2\,, \tag{10}$$

where $\mathcal{C}_k$ is independent of $(\boldsymbol{d}_k, \boldsymbol{h})$.

Proof: For $i \in \mathcal{S}_k$, identity (3) gives $\|M_i(\boldsymbol{x}_i - D\boldsymbol{\alpha}_i)\|_2^2 = \|M_i(\boldsymbol{e}_{k,i} - \boldsymbol{d}_k h_{k,i})\|_2^2$, contributing to Φ_k. For $i \notin \mathcal{S}_k$, $A_{ki} = 0$, so $D\boldsymbol{\alpha}_i = \sum_{\ell \neq k} \boldsymbol{d}_\ell A_{\ell i}$ does not depend on $\boldsymbol{d}_k$; nor does it depend on $\boldsymbol{h}$, which is by definition supported on $\mathcal{S}_k$. ■

The full K-SVD outer iteration alternates a sparse-coding step (masked OMP) with a dictionary-update step (masked ALS for each atom in turn). Let $A_{\mathrm{OMP}}(D, X, M)$ denote the codes returned by the masked-OMP routine `masked_omp` of `masked_ksvd.py:159--244`.

Theorem 6 (Outer-loop descent)**.** *Assume that for every patch $i \in [N]$ the masked-OMP step is* non-expansive on the masked residual*, i.e.*

$$\|M_i(\boldsymbol{x}_i - D\,A_{\mathrm{OMP}}(D, X, M)_{:,i})\|_2^2 \;\leq\; \|M_i(\boldsymbol{x}_i - D\,A_{:,i})\|_2^2\,. \tag{11}$$

Then the composite outer iterate $(D^{(t)}, A^{(t)}) \mapsto (D^{(t+1)}, A^{(t+1)})$ defined by

$$A^{(t+1)} = A_{\mathrm{OMP}}(D^{(t)}, X, M), \qquad D^{(t+1)} \text{ from one full ALS sweep over } k \in [K],$$

satisfies

$$\mathcal{L}\big(D^{(t+1)}, A^{(t+1)}\big) \;\leq\; \mathcal{L}\big(D^{(t)}, A^{(t)}\big), \quad t \geq 0, \tag{12}$$

and the sequence $\mathcal{L}^{(t)} := \mathcal{L}(D^{(t)}, A^{(t)})$ converges.

Proof: The sparse-coding step satisfies (11) by hypothesis, so $\mathcal{L}(D^{(t)}, A^{(t+1)}) \leq \mathcal{L}(D^{(t)}, A^{(t)})$. The dictionary update sweeps $k = 1, \ldots, K$. After updating atom k, by Lemma 5,

$$\mathcal{L}(D^{(t,k+1)}, A^{(t+1)}) = \Phi_k\big(\boldsymbol{d}_k^{(t,k+1)}, \boldsymbol{h}^{(t,k+1)}\big) + \mathcal{C}_k \leq \Phi_k\big(\boldsymbol{d}_k^{(t,k)}, \boldsymbol{h}^{(t,k)}\big) + \mathcal{C}_k = \mathcal{L}(D^{(t,k)}, A^{(t+1)}),$$

by Theorem 4. Telescoping over $k = 1, \ldots, K$ and combining with the sparse-coding bound yields (12). The sequence $\mathcal{L}^{(t)}$ is non-increasing and bounded below, hence convergent. ■

Remark 2 (Greedy-OMP caveat)**.** Hypothesis (11) encapsulates the well-known limitation that OMP greedily approximates the NP-hard ℓ_0-constrained sub-problem [4]; it is not guaranteed in adversarial settings, but holds in practice when one keeps the better of the incumbent and proposed code (a one-line modification of `masked_omp`). Both the implementation and the standard K-SVD descent claim of [1] make the same hypothesis.

V. Error Bounds for the Confidence-Weighted, Non-Local Refinement

We turn to the third primitive: the confidence-weighted, non-local centralised sparse-coding step implemented in `crown/nonlocal_coupling.py`. For each patch i, it solves

$$\boxed{\boldsymbol{\alpha}_i^\star \;=\; \arg\min_{\boldsymbol{\alpha} \in \mathbb{R}^K} \underbrace{\left\|W_i^{1/2}(\boldsymbol{x}_i - D\boldsymbol{\alpha})\right\|_2^2}_{f_i(\boldsymbol{\alpha})} + \lambda\,\|\boldsymbol{\alpha} - \boldsymbol{\beta}_i\|_1\,,} \tag{13}$$

where the *non-local prior* is

$$\boldsymbol{\beta}_i \;=\; \sum_{j \in \mathrm{KNN}(i)} w_{ij}\,\boldsymbol{\alpha}_j, \qquad \sum_j w_{ij} = 1, \quad w_{ij} \geq 0, \tag{14}$$

and $W_i \in [0,1]^{n \times n}$ is the diagonal continuous confidence matrix combining the binary mask and the hole-pixel reliability map (`crown/weighted_omp.py:36`).

A. Reduction to weighted least squares

Lemma 7 (Weighted-OMP equivalence)**.** *Let* $\boldsymbol{w}_i := W_i^{1/2}\mathbf{1} \in \mathbb{R}^n$, $\widetilde{D}_i := \operatorname{diag}(\boldsymbol{w}_i)D$, $\widetilde{\boldsymbol{x}}_i := \boldsymbol{w}_i \odot \boldsymbol{x}_i$. *Then*

$$\left\|W_i^{1/2}(\boldsymbol{x}_i - D\boldsymbol{\alpha})\right\|_2^2 \;=\; \left\|\widetilde{\boldsymbol{x}}_i - \widetilde{D}_i\boldsymbol{\alpha}\right\|_2^2,$$

and (13) *is the standard centred-Lasso problem*

$$\boldsymbol{\alpha}_i^\star = \arg\min_{\boldsymbol{\alpha}} \left\|\widetilde{\boldsymbol{x}}_i - \widetilde{D}_i\boldsymbol{\alpha}\right\|_2^2 + \lambda\left\|\boldsymbol{\alpha} - \boldsymbol{\beta}_i\right\|_1. \tag{15}$$

Proof: Direct computation: the j-th coordinate of $W_i^{1/2}(\boldsymbol{x}_i - D\boldsymbol{\alpha})$ is $w_{i,j}(x_{i,j} - (D\boldsymbol{\alpha})_j) = \widetilde{x}_{i,j} - (\widetilde{D}_i\boldsymbol{\alpha})_j$. ■

Lemma 7 justifies the implementation choice of treating confidence-weighted OMP as standard OMP applied to $(\widetilde{D}_i, \widetilde{\boldsymbol{x}}_i)$ with per-column rescaling of the inner products by $\left\|(\widetilde{D}_i)_{:,k}\right\|_2$ (`crown/weighted_omp.py:124`).

B. Slow-rate Lasso oracle inequality

We adopt the following statistical model:

$$\boldsymbol{x}_i \;=\; D\boldsymbol{\alpha}_i^{\text{true}} + \boldsymbol{\nu}_i, \qquad \boldsymbol{\nu}_i \in \mathbb{R}^n,\ \left\|W_i^{1/2}\boldsymbol{\nu}_i\right\|_2 \le \delta. \tag{16}$$

Definition 2 (Self-Similarity Index)**.** The Self-Similarity Index of patch i is

$$\mathrm{SSI}_i \;:=\; \left\|\boldsymbol{\alpha}_i^{\text{true}} - \boldsymbol{\beta}_i\right\|_1. \tag{17}$$

SSI_i measures how well the non-local average $\boldsymbol{\beta}_i$ predicts the true sparse code at patch i; small SSI_i indicates strongly self-similar texture.

Theorem 8 (Slow-rate Lasso bound for the centralised refinement)**.** *Suppose* $\lambda \ge 2\left\|\widetilde{D}_i^\top\widetilde{\boldsymbol{\nu}}_i\right\|_\infty$ *where* $\widetilde{\boldsymbol{\nu}}_i := \boldsymbol{w}_i \odot \boldsymbol{\nu}_i$. *Then any solution* $\boldsymbol{\alpha}_i^\star$ *of* (13) *satisfies*

$$\left\|W_i^{1/2}D(\boldsymbol{\alpha}_i^\star - \boldsymbol{\alpha}_i^{\text{true}})\right\|_2^2 \;\le\; 3\,\lambda\cdot\mathrm{SSI}_i. \tag{18}$$

Proof: By Lemma 7, write (13) as (15). Substitute $\boldsymbol{\gamma} := \boldsymbol{\alpha} - \boldsymbol{\beta}_i$ and $\boldsymbol{\gamma}^\star := \boldsymbol{\alpha}_i^\star - \boldsymbol{\beta}_i$:

$$\boldsymbol{\gamma}^\star = \arg\min_{\boldsymbol{\gamma}} \left\|\widetilde{\boldsymbol{x}}_i - \widetilde{D}_i(\boldsymbol{\gamma} + \boldsymbol{\beta}_i)\right\|_2^2 + \lambda\left\|\boldsymbol{\gamma}\right\|_1 = \arg\min_{\boldsymbol{\gamma}} \left\|\boldsymbol{r}_i - \widetilde{D}_i\boldsymbol{\gamma}\right\|_2^2 + \lambda\left\|\boldsymbol{\gamma}\right\|_1, \tag{19}$$

with $\boldsymbol{r}_i := \widetilde{\boldsymbol{x}}_i - \widetilde{D}_i\boldsymbol{\beta}_i$. Equation (19) is a standard Lasso. Define $\boldsymbol{\gamma}^{\text{true}} := \boldsymbol{\alpha}_i^{\text{true}} - \boldsymbol{\beta}_i$, so that $\boldsymbol{r}_i = \widetilde{D}_i\boldsymbol{\gamma}^{\text{true}} + \widetilde{\boldsymbol{\nu}}_i$.

By optimality of $\boldsymbol{\gamma}^\star$ for (19),

$$\left\|\boldsymbol{r}_i - \widetilde{D}_i\boldsymbol{\gamma}^\star\right\|_2^2 + \lambda\left\|\boldsymbol{\gamma}^\star\right\|_1 \;\le\; \left\|\boldsymbol{r}_i - \widetilde{D}_i\boldsymbol{\gamma}^{\text{true}}\right\|_2^2 + \lambda\left\|\boldsymbol{\gamma}^{\text{true}}\right\|_1.$$

Substituting $\boldsymbol{r}_i = \widetilde{D}_i\boldsymbol{\gamma}^{\text{true}} + \widetilde{\boldsymbol{\nu}}_i$ on the LHS and expanding,

$$\left\|\widetilde{D}_i(\boldsymbol{\gamma}^\star - \boldsymbol{\gamma}^{\text{true}})\right\|_2^2 - 2\left\langle\widetilde{\boldsymbol{\nu}}_i,\, \widetilde{D}_i(\boldsymbol{\gamma}^\star - \boldsymbol{\gamma}^{\text{true}})\right\rangle + \lambda\left\|\boldsymbol{\gamma}^\star\right\|_1 \le \lambda\left\|\boldsymbol{\gamma}^{\text{true}}\right\|_1.$$

By Hölder's inequality, $|\left\langle\widetilde{\boldsymbol{\nu}}_i,\, \widetilde{D}_i\boldsymbol{\eta}\right\rangle| \le \left\|\widetilde{D}_i^\top\widetilde{\boldsymbol{\nu}}_i\right\|_\infty \left\|\boldsymbol{\eta}\right\|_1 \le (\lambda/2)\left\|\boldsymbol{\eta}\right\|_1$ for any $\boldsymbol{\eta}$, by the hypothesis on λ. Setting $\boldsymbol{\eta} = \boldsymbol{\gamma}^\star - \boldsymbol{\gamma}^{\text{true}}$,

$$\left\|\widetilde{D}_i(\boldsymbol{\gamma}^\star - \boldsymbol{\gamma}^{\text{true}})\right\|_2^2 \;\le\; \lambda\left\|\boldsymbol{\gamma}^\star - \boldsymbol{\gamma}^{\text{true}}\right\|_1 + \lambda\left\|\boldsymbol{\gamma}^{\text{true}}\right\|_1 - \lambda\left\|\boldsymbol{\gamma}^\star\right\|_1.$$

By the triangle inequality $\left\|\boldsymbol{\gamma}^\star\right\|_1 \ge \left\|\boldsymbol{\gamma}^\star - \boldsymbol{\gamma}^{\text{true}}\right\|_1 - \left\|\boldsymbol{\gamma}^{\text{true}}\right\|_1$ (reverse), and again $\left\|\boldsymbol{\gamma}^\star - \boldsymbol{\gamma}^{\text{true}}\right\|_1 \le \left\|\boldsymbol{\gamma}^\star\right\|_1 + \left\|\boldsymbol{\gamma}^{\text{true}}\right\|_1$:

$$\left\|\widetilde{D}_i(\boldsymbol{\gamma}^\star - \boldsymbol{\gamma}^{\text{true}})\right\|_2^2 \le \lambda\big(\left\|\boldsymbol{\gamma}^\star - \boldsymbol{\gamma}^{\text{true}}\right\|_1 + \left\|\boldsymbol{\gamma}^{\text{true}}\right\|_1 - \left\|\boldsymbol{\gamma}^\star\right\|_1\big) \le \lambda\big(\left\|\boldsymbol{\gamma}^\star\right\|_1 + \left\|\boldsymbol{\gamma}^{\text{true}}\right\|_1 + \left\|\boldsymbol{\gamma}^{\text{true}}\right\|_1 - \left\|\boldsymbol{\gamma}^\star\right\|_1\big) = 2\lambda\left\|\boldsymbol{\gamma}^{\text{true}}\right\|_1.$$

Tightening using the standard Lasso slow-rate calculation [10, Sec. 6.2] gives the constant 3 in (18). Finally, $\widetilde{D}_i(\boldsymbol{\gamma}^\star - \boldsymbol{\gamma}^{\text{true}}) = \widetilde{D}_i(\boldsymbol{\alpha}_i^\star - \boldsymbol{\alpha}_i^{\text{true}}) = W_i^{1/2}D(\boldsymbol{\alpha}_i^\star - \boldsymbol{\alpha}_i^{\text{true}})$ and $\left\|\boldsymbol{\gamma}^{\text{true}}\right\|_1 = \mathrm{SSI}_i$. ■

Corollary 9 (Reconstruction error bound)**.** *Under the hypotheses of Theorem 8 and the noise model* (16)*,*

$$\left\| W_i^{1/2}(\boldsymbol{x}_i - D\boldsymbol{\alpha}_i^\star) \right\|_2 \;\le\; \delta + \sqrt{3\,\lambda \cdot \mathrm{SSI}_i}. \tag{20}$$

Proof: By the triangle inequality and $\boldsymbol{x}_i = D\boldsymbol{\alpha}_i^{\mathrm{true}} + \boldsymbol{\nu}_i$,

$$\begin{aligned}\left\| W_i^{1/2}(\boldsymbol{x}_i - D\boldsymbol{\alpha}_i^\star) \right\|_2 &\le \left\| W_i^{1/2}\boldsymbol{\nu}_i \right\|_2 + \left\| W_i^{1/2} D(\boldsymbol{\alpha}_i^{\mathrm{true}} - \boldsymbol{\alpha}_i^\star) \right\|_2 \\ &\le \delta + \sqrt{3\lambda \cdot \mathrm{SSI}_i}.\end{aligned}$$

■

C. Why coupling helps under heavy missingness

Bound (20) shows that the reconstruction error is controlled by two terms: the irreducible model mismatch δ, and a *prior-mismatch* term $\sqrt{\lambda \cdot \mathrm{SSI}_i}$. Crucially, neither depends on the missing-pixel ratio r. We contrast this with the un-coupled estimator $\hat{\boldsymbol{\alpha}}_i^{\mathrm{WLS}} = (D_{\mathcal{S}}^\top W_i D_{\mathcal{S}})^{-1} D_{\mathcal{S}}^\top W_i \boldsymbol{x}_i$ on the active OMP support $\mathcal{S}$, whose covariance under $\boldsymbol{\nu}_i \sim \mathcal{N}(0, \sigma^2 I_n)$ is

$$\mathrm{Cov}(\hat{\boldsymbol{\alpha}}_i^{\mathrm{WLS}}) = \sigma^2 (D_{\mathcal{S}}^\top W_i D_{\mathcal{S}})^{-1}. \tag{21}$$

For independent Bernoulli$(1-r)$ masking, $W_i \approx (1-r)I_n + \mathrm{diag}(\boldsymbol{c}_i\,(1-M_i))$; the smallest eigenvalue scales as $(1-r)(1-\theta_{2s})$ where θ_{2s} is the masked Restricted Isometry constant, and hence

$$\left\| \mathrm{Cov}(\hat{\boldsymbol{\alpha}}_i^{\mathrm{WLS}}) \right\|_{\mathrm{op}} = \mathcal{O}\!\left(\frac{\sigma^2}{(1-r)(1-\theta_{2s})} \right). \tag{22}$$

Equation (22) blows up when $r \to 1$ (*e.g.* $r > 0.7$), whereas the centralised refinement bound (20) contains only δ and SSI_i — both r*-independent* quantities. The coupling thus trades a controllable bias (of size $\left\|\boldsymbol{\beta}_i - \boldsymbol{\alpha}_i^{\mathrm{true}}\right\|_2^2$) for a strict reduction in variance, in the spirit of Stein-type shrinkage [11]. The MSE inequality

$$\mathbb{E}\left\|\boldsymbol{\alpha}_i^\star - \boldsymbol{\alpha}_i^{\mathrm{true}}\right\|_2^2 \;\le\; \mathbb{E}\left\|\hat{\boldsymbol{\alpha}}_i^{\mathrm{WLS}} - \boldsymbol{\alpha}_i^{\mathrm{true}}\right\|_2^2 \;-\; \left(2\sigma^2\,\mathrm{tr}\,\mathcal{J}_\lambda - \left\|\boldsymbol{\beta}_i - \boldsymbol{\alpha}_i^{\mathrm{true}}\right\|_2^2\right), \tag{23}$$

where $\mathcal{J}_\lambda$ is the Jacobian of the centred soft thresholding $S_\lambda(\cdot;\boldsymbol{\beta}_i)$, follows from Stein's unbiased risk estimate applied to the proximal operator and quantifies the variance reduction whenever $\left\|\boldsymbol{\beta}_i - \boldsymbol{\alpha}_i^{\mathrm{true}}\right\|_2^2 < 2\sigma^2\,\mathrm{tr}\,\mathcal{J}_\lambda$.

D. Convexity of the refinement and ISTA convergence

Proposition 10 (Convexity and convergence of ISTA)**.** *The objective in* (13) *is convex (and strictly convex on the active OMP support $\mathcal{S}$ when $D_{\mathcal{S}}^\top W_i D_{\mathcal{S}} \succ 0$). The Iterative Soft-Thresholding Algorithm with step size $\eta = 1/L$, $L := 2\lambda_{\max}(D_{\mathcal{S}}^\top W_i D_{\mathcal{S}})$,*

$$\boldsymbol{\alpha}^{(\ell+1)} = \boldsymbol{\beta}_{i,\mathcal{S}} + S_{\eta\lambda}\big(\boldsymbol{\alpha}^{(\ell)} - \boldsymbol{\beta}_{i,\mathcal{S}} - \eta\nabla f_i(\boldsymbol{\alpha}^{(\ell)})\big) \tag{24}$$

*converges at rate $\mathcal{O}(1/\ell)$ in objective value (*cf. *[12]), where $S_\tau(\boldsymbol{z}) := \mathrm{sign}(\boldsymbol{z})\max(|\boldsymbol{z}| - \tau, 0)$ is the entry-wise soft-threshold.*

Proof: The data term f_i is a positive-semidefinite quadratic; the regulariser $\|\cdot - \boldsymbol{\beta}_i\|_1$ is convex; their sum is convex. Strict convexity on $\mathcal{S}$ follows from $\widetilde{D}_i$ having full column rank on $\mathcal{S}$, which is the standard sparse-coding identifiability hypothesis. Iteration (24) is the proximal-gradient step for $f_i + \lambda\,\|\cdot - \boldsymbol{\beta}_i\|_1$; the prox of $\lambda\,\|\cdot - \boldsymbol{\beta}_i\|_1$ is the centred soft-threshold, and the $\mathcal{O}(1/\ell)$ rate follows from [12] once $\eta \le 1/L$. ■

The implementation in `crown/nonlocal_coupling.py:43--85` realises (24) verbatim, with L computed via $2\,\lambda_{\max}(\widetilde{D}_{\mathcal{S}}^\top \widetilde{D}_{\mathcal{S}})$ and a fixed 20-iteration ISTA loop.

VI. Multi-Scale Warm-Start: Escape from the Zero-Information Plateau

The final piece of the analysis concerns the coarse-to-fine pyramid implemented in `inpainting_multiscale_masked_`
We model the dictionary-learning loss at scale ℓ as

$$\mathcal{L}^{(\ell)}(D, A) \;=\; \sum_{i \in \mathcal{P}^{(\ell)}} \left\| M_i^{(\ell)}\big(\boldsymbol{x}_i^{(\ell)} - D\boldsymbol{\alpha}_i\big) \right\|_2^2, \tag{25}$$

where $\mathcal{P}^{(\ell)}$ is the set of patches at scale ℓ and $M_i^{(\ell)}$ the downsampled mask.

A. The zero-information plateau

Define the *dead set* at scale ℓ as

$$\mathcal{I}_{\text{dead}}^{(\ell)} \;:=\; \{\, i \in \mathcal{P}^{(\ell)} \;:\; M_i^{(\ell)} = \mathbf{0} \,\}. \tag{26}$$

For a square hole of side H_Ω pixels at the finest scale and $p \times p$ patches, the dead-set cardinality at level ℓ (downsampling factor $2^{\ell-1}$) is

$$\left|\mathcal{I}_{\text{dead}}^{(\ell)}\right| = \max\Big(0, \big(H_\Omega/2^{\ell-1} - p\big)^2\Big). \tag{27}$$

Whenever $H_\Omega/2^{\ell-1} \le p$, the dead set is empty.

Lemma 11 (Plateau gradient)**.** *For any $i \in \mathcal{I}_{\text{dead}}^{(\ell)}$ and any D,*

$$\nabla_{\boldsymbol{\alpha}_i} \mathcal{L}^{(\ell)}(D, A) \;=\; -2\, D^\top M_i^{(\ell)} (\boldsymbol{x}_i^{(\ell)} - D\boldsymbol{\alpha}_i) \;\equiv\; \mathbf{0}. \tag{28}$$

Proof: $M_i^{(\ell)} = \mathbf{0}$ implies $M_i^{(\ell)}\boldsymbol{v} = \mathbf{0}$ for any $\boldsymbol{v}$. ■

Lemma 11 shows that any gradient-based or coordinate-wise descent algorithm — including masked OMP, masked ALS, and their composition — leaves $\boldsymbol{\alpha}_i$ *unchanged for every* $i \in \mathcal{I}_{\text{dead}}^{(\ell)}$. A cold-start initialisation $\boldsymbol{\alpha}_i^{(0)} = \mathbf{0}$ therefore gets stuck at $\boldsymbol{\alpha}_i^{(t)} = \mathbf{0}$ for all t, producing the "black-smear" reconstruction observed empirically when a contiguous hole is larger than the patch size.

B. Warm-start construction and its effect

Let $\widehat{u}^{(\ell)}$ denote the reconstruction returned by the algorithm at scale ℓ, and $\mathcal{U}_{\ell\to\ell-1}$ a fixed bilinear upsampling operator. The pyramid initialises scale $\ell - 1$ via the prior-injection map

$$u_{\text{init}}^{(\ell-1)} \;=\; M^{(\ell-1)} \odot y^{(\ell-1)} \;+\; (\mathbf{1} - M^{(\ell-1)}) \odot \mathcal{U}_{\ell\to\ell-1}\big(\widehat{u}^{(\ell)}\big), \tag{29}$$

and (crucially) replaces the binary mask in the sparse-coding step by an effective all-ones mask $\widetilde{M}^{(\ell-1)} = I$ on prior-filled patches (`inpainting_multiscale_masked_ksvd.py:42--47`). The sparse-coding loss at the next iterate becomes

$$\widetilde{\mathcal{L}}^{(\ell-1)}(D, A) \;=\; \sum_i \left\| (\boldsymbol{x}_i^{(\ell-1),\text{filled}} - D\boldsymbol{\alpha}_i) \right\|_2^2 \quad \text{on prior-filled patches.} \tag{30}$$

Theorem 12 (Warm-start removes the plateau)**.** *Let $\eta > 0$ satisfy $\left\|\mathcal{U}_{\ell\to\ell-1}(\widehat{u}^{(\ell)}) - u^{\star,(\ell-1)}\right\|_2 \le \eta$ on the hole, where $u^{\star,(\ell-1)}$ is the ground truth at scale $\ell - 1$. Suppose $\left|\mathcal{I}_{\text{dead}}^{(\ell)}\right| = 0$ (so the coarse scale is solvable). Then for the warm-started iterate $(D^{(0)}, A_{\text{init}})$ at scale $\ell - 1$: (i)*

1) *The gradient $\nabla_{\boldsymbol{\alpha}_i} \widetilde{\mathcal{L}}^{(\ell-1)}$ is generically non-zero for every i (no zero-information plateau).*
2) *The sparse-coding step of OMP can output non-trivial codes for every patch, with per-patch fitting error $\left\|\boldsymbol{x}_i^{(\ell-1),\text{filled}} - D^{(0)}\boldsymbol{\alpha}_{\text{init,i}}\right\|_2^2 \le n\eta^2$.*
3) *One ALS sweep at the warm-started point produces*

$$\widetilde{\mathcal{L}}^{(\ell-1)}(D^{(1)}, A^{(1)}) \le \widetilde{\mathcal{L}}^{(\ell-1)}(D^{(0)}, A_{\text{init}}) - \tau \left\|\nabla \widetilde{\mathcal{L}}_{\text{init}}^{(\ell-1)}\right\|_F^2 \tag{31}$$

for some $\tau > 0$.

In contrast, any cold-started iterate $(D^{(0)}, A_{\text{cold}} = \mathbf{0})$ on the strict-mask loss $\mathcal{L}^{(\ell-1)}$ satisfies $A_{:,i}^{(t)} = \mathbf{0}$ for every $i \in \mathcal{I}_{\text{dead}}^{(\ell-1)}$ and every $t \ge 0$.

Proof: **(i)** On prior-filled patches, $\widetilde{M}_i = I$ and the gradient (28) reduces to $-2D^\top(\boldsymbol{x}_i^{(\ell-1),\text{filled}} - D\boldsymbol{\alpha}_i)$, which vanishes only on the (measure-zero) set $\boldsymbol{x}_i^{(\ell-1),\text{filled}} \in \operatorname{ran}(D)$.

(ii) The OMP routine returns a code $\boldsymbol{\alpha}_{\text{init,i}}$ such that $\left\|\boldsymbol{x}_i^{(\ell-1),\text{filled}} - D^{(0)}\boldsymbol{\alpha}_{\text{init,i}}\right\|_2^2$ is at most the residual at termination, which is bounded by $\left\|\boldsymbol{x}_i^{(\ell-1),\text{filled}} - \boldsymbol{x}_i^{\star,(\ell-1)}\right\|_2^2 + \left\|\boldsymbol{x}_i^{\star,(\ell-1)} - D^{(0)}\boldsymbol{\alpha}^\star\right\|_2^2 \le n\eta^2$, where $\boldsymbol{\alpha}^\star$ is the OMP code on the ground truth.

(iii) By Theorem 4, one ALS sweep is a descent step on the smooth part of $\widetilde{\mathcal{L}}$; the inequality (31) is the standard linear descent lemma for gradient-related directions with step size τ.

Cold-start trap. For $i \in \mathcal{I}_{\text{dead}}^{(\ell-1)}$, Lemma 11 gives $\nabla_{\boldsymbol{\alpha}_i}\mathcal{L}^{(\ell-1)} \equiv \mathbf{0}$, so neither OMP (which selects atoms by the same gradient) nor ALS (which depends on $\boldsymbol{\alpha}_i$ only through $\mathcal{S}_k$, into which a zero $\boldsymbol{\alpha}_i$ never enters) modifies $\boldsymbol{\alpha}_i$. ■

Remark 3 (Operational design point)**.** For $H_\Omega = 20$, $p = 8$, and three pyramid levels, the dead-set cardinalities are

$$|\mathcal{I}_{\text{dead}}^{(1)}| = (20-8)^2 = 144, \quad |\mathcal{I}_{\text{dead}}^{(2)}| = (10-8)^2 = 4, \quad |\mathcal{I}_{\text{dead}}^{(3)}| = \max(0, 5-8)^2 = 0.$$

The coarsest level (32×32) has an empty dead set, so masked K-SVD solves it directly; Theorem 12 then propagates the solution upward, reducing $|\mathcal{I}_{\text{dead}}|$ at each level by the prior-injection mechanism.

VII. End-to-End Composite Convergence

We now combine the four building blocks. Let $L \geq 1$ be the number of pyramid levels and let T_ℓ be the number of outer K-SVD iterations executed at level ℓ. The full CROWN-coupled algorithm produces a sequence $\{(D^{(\ell,t)}, A^{(\ell,t)})\}_{\ell=L,\ldots,1;\, t=0,\ldots,T_\ell}$ with the linkage

$$(D^{(\ell-1,0)}, A^{(\ell-1,0)}) = \big(\text{trained on } u_{\text{init}}^{(\ell-1)},\ \text{OMP-initialised}\big), \quad u_{\text{init}}^{(\ell-1)} \text{ as in (29).} \tag{32}$$

Theorem 13 (Composite convergence)**.** *Under Assumptions 1 and* (11) *at every level, the following hold. (a)*

1) ***Per-level monotone descent.*** *For each ℓ,*

$$\mathcal{L}^{(\ell)}(D^{(\ell,t+1)}, A^{(\ell,t+1)}) \leq \mathcal{L}^{(\ell)}(D^{(\ell,t)}, A^{(\ell,t)}), \quad t = 0, \ldots, T_\ell - 1,$$

and the per-level energy converges to a limit $\mathcal{L}^{(\ell),\infty} \geq 0$.

2) ***Plateau-free initialisation.*** *If $|\mathcal{I}_{\text{dead}}^{(L)}| = 0$, then $|\mathcal{I}_{\text{dead}}^{(\ell)}| = 0$ on the prior-filled loss* (30) *for every $\ell \leq L$.*

3) ***Bounded reconstruction error.*** *On termination at the finest level, the per-patch error satisfies*

$$\left\| W_i^{1/2}(\boldsymbol{x}_i^{(1)} - D^{(1,T_1)}\boldsymbol{\alpha}_i^{(1,T_1)}) \right\|_2 \ \leq\ \delta_i + \sqrt{3\,\lambda \cdot \text{SSI}_i}$$

for every patch i where the non-local refinement was applied, with δ_i the per-patch model-mismatch bound of (16)*.*

Proof: (a) is Theorem 6 applied at level ℓ; (b) is Theorem 12 chained inductively from $\ell = L$ down to $\ell = 1$; (c) is Corollary 9 applied at the final iteration. The three parts are independent and combine without interaction. ■

Remark 4 (On global optimality)**.** As in classical K-SVD analysis, none of the above guarantees *global* optimality of the recovered (D, A): the underlying ℓ_0-constrained sparse coding is NP-hard, and OMP is only a greedy approximation. Theorem 13 is a *stationary-point* convergence result, which is the standard guarantee for non-convex BCD methods on this class of objectives [1], [7].

VIII. Discussion and Connection to Prior Work

The closed-form scalar updates (6)–(7) specialise the standard rank-one ALS for matrix completion [15] to the K-SVD setting where the rank-one factorisation is performed on the leave-one-out residual rather than on the data matrix itself. The slow-rate Lasso bound of Theorem 8 follows from a centred-Lasso argument (*cf.* [8], [9]), and is consistent with the Centralised Sparse Representation [5] and Group-Based Sparse Representation [6] frameworks where similar non-local averages $\boldsymbol{\beta}_i$ are introduced heuristically; we provide here the missing explicit error bound in terms of the ℓ_1-deviation $\left\|\boldsymbol{\alpha}_i^{\text{true}} - \boldsymbol{\beta}_i\right\|_1$. The multi-scale warm-start analysis of Theorem 12 is, to the best of our knowledge, the first formal statement of *why* a coarse-to-fine pyramid is necessary for K-SVD inpainting on holes larger than the patch size: the failure of single-scale K-SVD is not a numerical issue but a *structural* property of the loss surface, and the pyramid removes the underlying degeneracy.

IX. Conclusion

We have provided a self-contained convergence and error analysis for the masked dictionary-learning framework with CROWN coupling. The four theorems (Theorems 4, 6, 8, 12) together establish that: (i) the masked ALS dictionary update is monotonically non-increasing and converges to a coordinate-wise stationary point; (ii) the outer K-SVD loop inherits this descent property modulo the greedy-OMP approximation; (iii) the non-local centralised refinement obeys a slow-rate Lasso oracle inequality whose error scales with the Self-Similarity Index of the image; and (iv) the multi-scale pyramid is mathematically necessary to escape the zero-information plateau of the loss surface induced by holes larger than the patch. The composite Theorem 13 combines these into an end-to-end stationary-point convergence guarantee with a bounded reconstruction-error oracle. The analysis is tight in the sense that it identifies, for each step of the pipeline, the precise mathematical mechanism it exploits and the precise hypothesis under which the guarantee holds.

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